Jaime Jimenez Arevalo

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Education

Bachelor of Science in Mathematics Expected July 2027

– California State University, Dominguez Hills (CSUDH), Carson, California.

– Relevant coursework: Linear Algebra, Complex Analysis, and Abstract Algebra.

Associate of Science in Mathematics August 2022 – July 2025

– Long Beach City College (LBCC), Long Beach, California.

– Relevant coursework: Calculus, Linear Algebra, and Differential Equations.

Work Experience

Tutor/Supplemental Instructor August 2023 – Present

– Long Beach City College Math Success Center, Long Beach, California.

– Facilitated academic support workshops within the Math Success Center, enhancing student comprehension and exam readiness across multiple mathematics subjects.

– Partnered with professors to assist students during lectures and led supplemental review sessions outside of class to reinforce core course concepts.

– Conducted in-center tutoring across mathematics subjects ranging from entry-level courses to advanced topics such as Differential Equations and Linear Algebra.

– Promoted Math Success Center resources through classroom presentations and campus outreach events, engaging students and distributing informational materials.

Tutor and Residential Assistant June 2019 – July 2019

– TEF/TRiO Programs, Los Angeles, California.

– Collaborated with faculty to deliver academic readiness and personal development workshops for college-bound high school students.

– Supported program leadership by implementing an online calculus curriculum, providing tutoring, and advising students on their academic progress.

Research Experience

Undergraduate Researcher June 15, 2026 – July 24, 2026

NSF REU Site: Combinatorics and Coding Theory in the Tropics, University of Puerto Rico in Ponce.

– Conducted research exploring the mathematical structure of Affine Grassman Codes under the supervision of Dr. Fernando Piñero Gonzalez.

– Participated in workshops to develop technical skills in LaTeX, research techniques, and coding methods.

– Collaborated with undergraduate researchers to develop an understanding of the assigned project and produce logically sound proofs to defend the material.

Student Researcher November 2025 – Present

California State University, Dominguez Hills, Carson, California.

- Conducted undergraduate research exploring the mathematical structure of Liar's Bingo and its connections to error-correcting codes and coding theory.
- Curated a poster presentation for the CSUDH Student Research Conference, securing second place in the Physical and Mathematical Sciences Undergraduate Poster Presentations category.
- Developed Python simulations to model strip construction, introduce errors, and extend the game to different bases and strip sizes.

Presentations

Decoding Polar Affine Grassmann Codes July 2026

Ohio State University, Young Mathematicians Conference

The Intersection Between Liar's Bingo and Coding Theory February 2026

California State University, Dominguez Hills, Student Research Conference

Community Involvement

CSUDH Math Club – Secretary November 2025 – Present

- Promoted club activities and fostered a collaborative club environment.
- Managed club records, took meeting minutes, and prepared agendas for club meetings.

LBCC Math Club – Founder and President November 2023 – December 2024

- Re-established the Math Club at Long Beach City College to engage students and foster inclusion.
- Wrote grants to funding opportunities to secure financial support for club activities.

Awards and Honors

LSAMP Research Grant $$,$$$ Setpember 2025 - present

Constructed abstract to get funding for my research at CSUDH. Research based on the Intersection between Liar's Bingo and Coding Theory.

Second Place, Undergraduate Poster Presentation Spring 2026

Awarded at the California State University, Dominguez Hills Student Research Conference for a poster presentation in Physical and Mathematical Sciences Undergraduate Poster Presentations section. Research focused on Liar's Bingo, error-correcting codes, and coding theory.

LBCC Auxiliary Grant, $1,059 November 2024

Secured grants to fund club operations such as tutoring sessions, guest

Skills

Programming Languages	Java, C++, Python
Computer Software/Frameworks	Adobe Photoshop, LaTeX, Microsoft Office, OBS
Languages	English, Spanish